

WILKES-BARRE SCRANTON, PA, USA

WMO: 725130

Lat: 41.334N

Lon: 75.727W

Elev: 930

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14777

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.2	(c) 8.9	(d) -7.7	(e) 3.8	(f) 6.4	(g) -3.2	(h) 4.8	(i) 11.2	(j) 24.2	(k) 30.9	(l) 21.2	(m) 28.5	(n) 8.3	(o) 230	(p) 0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	89.2	72.1	86.5	70.7	83.8	69.2	74.9	84.2	73.3	82.0	72.0	80.0	9.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	122.9	79.2	70.6	116.8	77.5	69.3	111.5	76.3	38.9	84.3	37.5	82.3	36.2	79.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	17.8	15.9	-1.5	93.3	6.1	2.8	-5.9	95.3	-9.5	97.0	-13.0	98.6	-17.5	100.7
			-2.5	77.9	5.9	2.2	-6.8	79.5	-10.3	80.8	-13.6	82.0	-17.9	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	26.7	29.5	37.5	49.5	59.9	68.0	72.5	70.7	63.9	52.5	41.9	32.6		
	DBStd	18.01	11.10	9.96	10.18	9.23	7.86	6.28	5.25	5.18	7.07	8.27	8.96	9.21		
	HDD50	2730	725	577	410	119	9	0	0	0	1	70	274	545		
	HDD65	5965	1187	995	855	472	200	42	5	11	104	395	694	1005		
	CDD50	2929	3	2	21	104	317	539	699	642	419	148	30	5		
	CDD65	688	0	0	1	7	43	131	239	188	71	8	0	0		
	CDH74	5559	0	0	8	100	435	1048	2002	1406	517	42	1	0		
	CDH80	1607	0	0	2	23	116	280	657	392	133	4	0	0		
(14) Wind	WSAvg	6.5	7.3	7.6	7.9	7.7	6.4	5.7	5.4	5.0	5.4	5.9	6.6	6.9		
(15) Precipitation	PrecAvg	37.2	2.3	2.1	2.6	3.1	3.5	3.9	3.7	3.5	3.8	3.1	3.0	2.6		
	PrecMax	61.1	6.5	8.1	5.4	9.6	8.0	9.1	7.2	10.6	10.4	8.1	7.7	6.6		
	PrecMin	26.2	0.4	0.3	0.5	0.9	0.8	0.3	1.0	1.2	0.7	0.0	0.8	0.9		
	PrecStd	7.4	1.3	1.3	1.1	1.5	1.5	1.9	1.7	1.7	2.5	1.8	1.4	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.3	61.5	72.6	83.5	87.5	90.3	93.2	91.7	89.0	78.9	69.3	63.2		
		MCWB	56.9	53.8	58.1	62.1	69.1	73.5	73.8	73.5	70.2	66.5	58.4	57.6		
	2%	DB	54.0	53.7	64.8	76.2	83.4	86.6	89.5	87.6	83.7	73.8	64.0	55.7		
		MCWB	49.3	46.3	52.4	58.8	66.6	71.4	72.8	71.1	68.3	63.5	56.7	51.6		
	5%	DB	47.1	48.1	58.9	70.4	79.0	83.2	86.7	84.2	80.0	70.2	60.1	50.2		
		MCWB	43.0	41.8	49.4	56.4	64.6	69.4	71.4	69.7	67.2	61.6	53.4	46.3		
	10%	DB	41.7	43.3	53.4	65.5	74.5	80.2	83.7	81.4	76.1	66.4	56.1	45.6		
		MCWB	38.2	38.3	45.9	53.3	62.3	67.5	70.0	68.5	65.6	58.9	50.0	41.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.2	55.6	59.6	65.8	72.2	75.8	77.9	76.7	74.1	69.5	62.1	58.2		
		MCDB	61.1	59.6	69.7	75.7	82.9	85.6	88.5	86.2	82.8	73.7	66.4	62.4		
	2%	WB	50.8	47.6	54.9	62.0	69.3	73.5	75.2	74.1	71.4	66.3	58.4	52.2		
		MCDB	53.3	52.2	62.1	72.4	79.2	82.9	85.0	82.5	79.3	71.7	63.2	54.7		
	5%	WB	43.6	42.5	50.8	58.7	67.0	71.8	73.5	72.5	69.5	63.3	54.5	46.9		
		MCDB	46.9	47.1	57.7	67.8	76.0	79.8	82.4	80.3	75.7	68.4	59.0	50.0		
	10%	WB	38.8	38.7	46.1	55.1	64.4	70.0	72.1	71.0	67.9	59.5	50.9	41.8		
		MCDB	41.4	43.0	53.2	64.1	71.9	77.6	80.3	78.2	73.9	65.1	55.0	45.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.9	17.0	19.9	20.3	19.3	19.9	19.2	19.1	17.7	14.9	12.1		
		MCDBR	19.1	20.9	24.6	27.4	25.8	23.0	22.6	22.1	23.5	22.3	20.2	17.1		
		MCWBR	16.8	16.1	15.4	15.0	12.0	10.4	8.9	9.2	10.6	12.5	14.1	14.6		
	5% WB	MCDBR	18.2	19.1	22.0	23.7	22.1	19.3	18.9	18.3	18.1	18.4	17.5	16.5		
		MCWBR	16.8	16.4	16.1	15.1	12.0	10.1	8.6	8.5	9.6	12.2	14.1	15.5		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.326	0.351	0.372	0.409	0.462	0.467	0.455	0.374	0.362	0.316	0.298		
	taud		2.401	2.317	2.350	2.345	2.273	2.152	2.152	2.174	2.377	2.414	2.504	2.481		
	Ebn at Noon		270	279	284	285	276	260	257	257	273	263	264	263		
	Edh at Noon		26	32	35	38	42	48	47	45	34	30	23	22		
(44) All-Sky Solar Radiation	RadAvg		521	795	1102	1430	1629	1755	1817	1583	1298	870	591	427		
	RadStd		38	97	96	155	170	163	111	98	163	106	68	53		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-302	+279	+135
(47) Regional (53 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	+113

Nomenclature: See separate page